Shiftless Collatz Model and the Ternary Billiards: A Geometric Interpretation of the Trajectory

Hiroshi Harada – March 27, 2026

1. Overview

This report is based on the rewrite system introduced in the previous study *"Shiftless Collatz Model: Binary Interference in the 3n+LSB Trajectory."* The Shiftless Collatz Model is mathematically equivalent to the standard Collatz iteration, but it eliminates the explicit "divide-by-two" operation, thereby ensuring no loss of information in the transformation process. In this study, the Shiftless Collatz trajectory is interpreted as a billiard motion on a $\log_3$-lattice plane, revealing the hidden physical structure of *warp* and *target convergence* within the chaotic dynamics.

The model is defined by the recurrence:

$$n_{k+1} = 3n_k + 2^{v_2(n_k)}$$

2. The Stage: The Ternary Billiards

To analyze the trajectory, we introduce the following coordinate system:

- Horizontal axis (time): step index k
- Vertical axis (lattice height): $y = \log_3(n_k)$
- Inertia Line: $y = k + \log_3(n_0)$

  The straight line corresponding to pure geometric growth by a factor of 3
- Jackpot Lines: $y = M\log_3(2)$

  These horizontal slits represent the powers of two, 2^M. When the trajectory reaches one of these lines, the process terminates—this is the "jackpot."

3. Dynamics of the Trajectory

The classical $3n + 1$ behavior appears on this billiard table as a combination of inertia and warp:

- Inertia: steady upward motion due to the $3n$ shift.
- Warp: a small upward boost caused by the LSB contribution $2^{v_2(n_k)}$.

The increment of the trajectory in the vertical direction is:

$$\Delta y = \log_3(n_{k+1}) - \log_3(n_k) = 1 + \log_3\left(1 + \frac{2^{v_2(n_k)}}{3n_k}\right)$$

The second term is the source of the warp energy that bends the trajectory away from the inertia line.

4. Case Study: The Persistence of Seed 27

Starting from 27, the trajectory undergoes a long journey of 41 steps before finally reaching the jackpot at 2^{64}.

The behavior can be divided into three phases:

- Early phase: The LSB contribution is relatively large, causing strong deviations from the inertia line and repeated warp oscillations.
- Middle phase: The trajectory repeatedly grazes the jackpot lines—near misses—yet subtle binary interference forces it to overshoot and remain on the board.
- Final phase: A long accumulation of internal binary "1"s triggers a massive carry event, producing a local growth rate r that peaks at 3.2. This final surge propels the trajectory cleanly into the 2^{64} jackpot line.

$$r = \frac{n_{k+1}}{n_k}$$

5. Conclusion

The ternary billiard model transforms the Collatz problem from a numerical iteration into a geometric ballistic prediction. This viewpoint provides an intuitive explanation for: "why certain numbers undergo extremely long trajectories," and "why others suddenly converge." The answer lies in the geometric distortion caused by binary interference—a physical intuition hidden within the arithmetic of the Collatz map.

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